

Chromosomes

A cell needs to make a copy of its DNA before it reproduces by dividing in two. DNA is packed into dense structures called chromosomes, found inside the nucleus (core) of the cell. Each chromosome gets copied before the cell can divide. Humans have 23 pairs of chromosomes in each cell.

A point on a chromosome called a centromere holds the chromosome and its copy together until the cell has divided.

Key (bases)

- Guanine
- Cytosine
- Thymine
- Adenine

Bases pair up like rungs on a ladder.



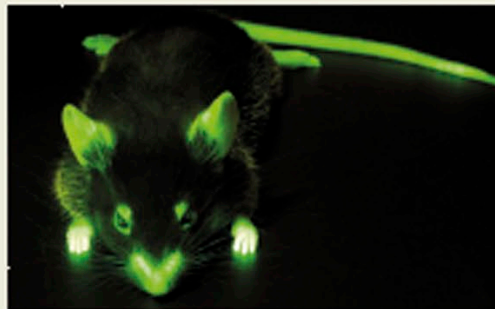
Genomes

A genome is the collection of all the genes of a species. Scientists are at work mapping entire genomes by identifying all the base pairs that make up the genes. The human genome has more than 22,000 genes containing about 3.2 billion base pairs.

Application of genetics

Genetic engineering

The DNA of organisms can be altered by using enzymes to cut out small pieces of DNA from one species and insert some of its genes into another one. This can give the genetically modified (GM) species useful attributes, such as increasing its nutritional content or making it more resistant to pests.



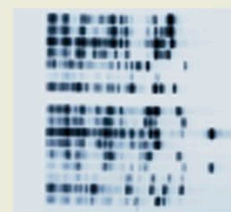
This GM mouse glows in the dark because of a gene for a fluorescent protein.

Tackling disease

Some diseases, such as cystic fibrosis and color blindness, are inherited. Genetic screening enables doctors to assess whether a patient may be vulnerable to a disease. Gene therapy is an emerging field of medical treatment. In it, a gene that causes a disease is either removed or replaced with a healthy gene.

DNA fingerprinting

With the exception of identical twins, each person's genes (their genetic fingerprint) are unique. DNA fingerprinting helps to identify relationships between family members. Law enforcement agencies use it to identify people from traces of their DNA left behind in hair, skin cells, or other body samples at crime scenes.



Each row is the DNA fingerprint of different people in a family.

1985

British geneticist Alec Jeffreys and his team at Leicester University, UK, pioneered DNA profiling, a process in which small parts of different people's DNA are compared to identify them.

1999

The first complete human chromosome (Chromosome 22) was mapped as part of the Human Genome Project.

2002

The mouse was the first mammal to have its full genome sequence mapped. It consisted of 3.48 billion bases.



2008

The 1000 Genomes Project was started. Its aim was to map the genomes of more than 1,000 people to learn about the variations in their genes.